

Supplementary Materials for GlyTwin

I. TRAINING CURVES, CALIBRATION AND PER-CLASS PERFORMANCE

A. MLP Classifier

Figures 1a-1d shows the training curves for the MLP classifier.

Table I shows the per-class performance of the MLP classifier.

Figure 2 is the confusion matrix for the classifier.

Class imbalance: Class imbalance was addressed by applying inverse-frequency class weighting during model training, which ensured that minority-class samples contributed proportionally more to the loss function. Specifically, class weights were computed using the balanced scheme in `sklearn`, where each class weight is inversely proportional to its prevalence in the training data, and these weights were added into the training objective to mitigate bias toward the majority class.

B. XGBoost simulator

The XGBoost simulator was calibrated using a grid search method in which key hyperparameters were systematically

TABLE I: Per-class classification results for the classifier

Class	Precision	Recall	F1-score	Support
normoglycemia	0.80	0.92	0.86	194
hyperglycemia	0.88	0.70	0.78	150
Accuracy			0.83	344
Macro Avg	0.84	0.81	0.82	344
Weighted Avg	0.83	0.83	0.82	344

varied within predefined ranges. The grid explored maximum tree depth values in the range [1, 12], learning rates spanning [0.001, 0.99], and numbers of estimators across [1, 100]. The search was optimized using the F1-score. Table II shows class-wise performance for the XGBoost simulation.

TABLE II: Per-class performance metrics for the XGBoost simulator

Class	Precision	Recall	F1-score
Class 0 (normoglycemia)	0.8088	0.9072	0.8552
Class 1 (hyperglycemia)	0.8475	0.8067	0.8266

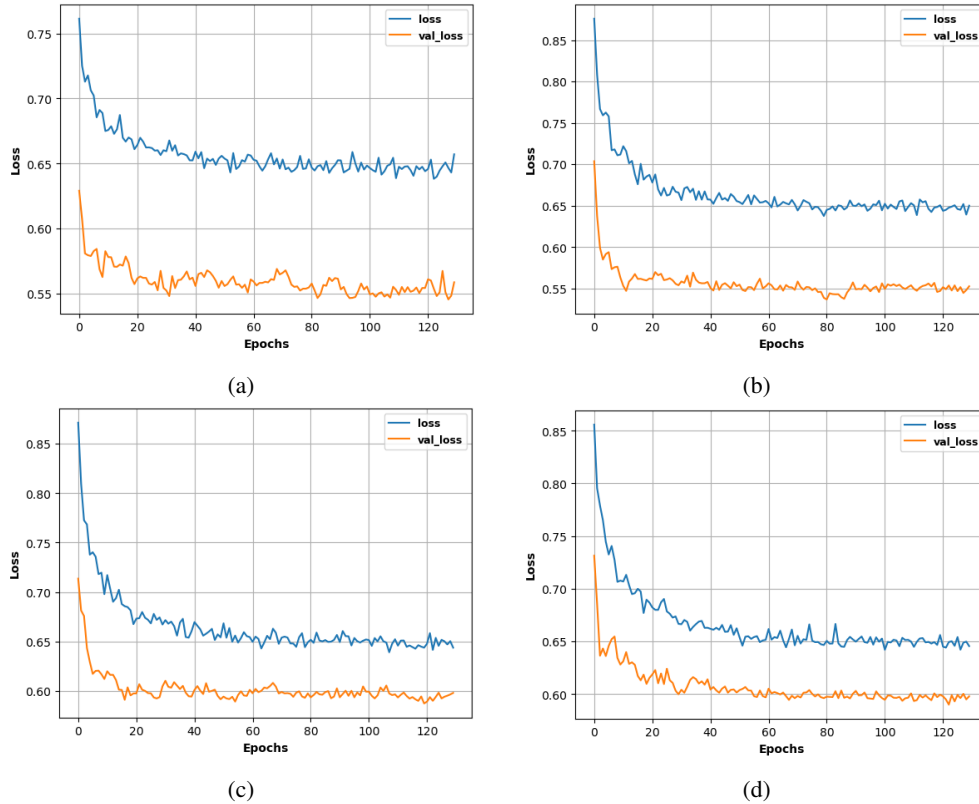


Fig. 1: Training curves for different seed values.

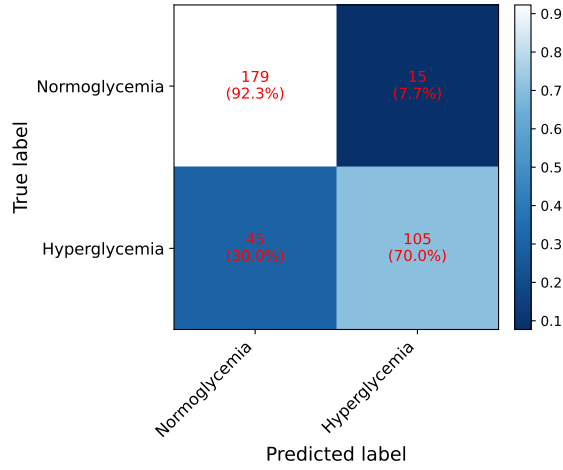


Fig. 2: Confusion matrix for the classifier.

C. NN test

This NN test uses *Euclidean distance* (Minkowski with $p = 2$) to find the 7 nearest neighbors, gives them *uniform* (equal) voting weights, automatically selects the neighbor search algorithm with a *leaf size* of 30 for efficiency. Table III shows class-wise performance for the NN-test.

TABLE III: Per-class classification results for NN-test

Class	Precision	Recall	F1-score	Support
normoglycemia	0.83	0.88	0.85	194
hyperglycemia	0.84	0.78	0.81	150
Accuracy			0.83	344
Macro Avg	0.84	0.83	0.83	344
Weighted Avg	0.84	0.84	0.84	344

II. BASELINE CONFIGURATION

Table IV contains configurations for baseline methods: DiCE, NICE, CFNOW and Optbinning.

TABLE IV: Configuration of different baseline methods

Component	DiCE	NICE	CFNOW	Optioning
Total CFs per factual	1	-	-	n_cf = 1
Desired class	opposite	-	-	-
Distance metric	-	HEOM	-	-
Optimization objective	-	proximity	-	-
Justified counterfactuals	-	True	-	-
Time limit	-	-	10 s	-
Number of CFs generated	-	-	1	1
Base classifier (model)	-	-	-	LogisticRegression (C = 0.0001)
Model accuracy	-	-	-	80.52%
Scorecard construction	-	-	-	Scorecard(binning_process = binning_process, estimator = estimator, scaling_method = "min_max", scaling_method_params = {0, 100}, reverse_scorecard = True)
Features allowed to vary / Actionable features	CarbSize, TotalBolus, del_t, PreMeal_BGL	-	-	CarbSize, TotalBolus, del_t, PreMeal_BGL
Permitted range source	Patient-specific min/max	-	-	-
Permitted range (per feature)	[min, max] on matching subset	-	-	-
Weights / Hyperparameters	proximity_weight = 0.5	Library defaults	-	-
Maximum number of changes	-	-	-	max_changes = 4
Hard constraints	-	-	-	diversity_values
Other hyperparameters	Library defaults	Library defaults	-	-

III. LOOPINSIGHT1 SIMULATOR

Below is an example of validating an intervention using the simulator. All the details from the interventions (Table V) are fed to the simulator as input and the resulting glycemic response is monitored (Figure 3).

TABLE V: A counterfactual intervention to be fed to the LoopInsighT1 simulator

Feature	Value
Age	76
Gender	F
Ethnicity	White
A1C	6.8
Carb size	36
Total bolus	4.46
Δt	-20
Mode	regular
Total basal	0.5485
Pre-meal BGL slope	2.83
Pre-meal BGL	113

IV. VIDEO LINKS

Video description of how the basal rates were extracted using OCR: tinyurl.com/4e2t6auv

Video description of how the device modes were extracted using a coordinate system: tinyurl.com/494jhu9e

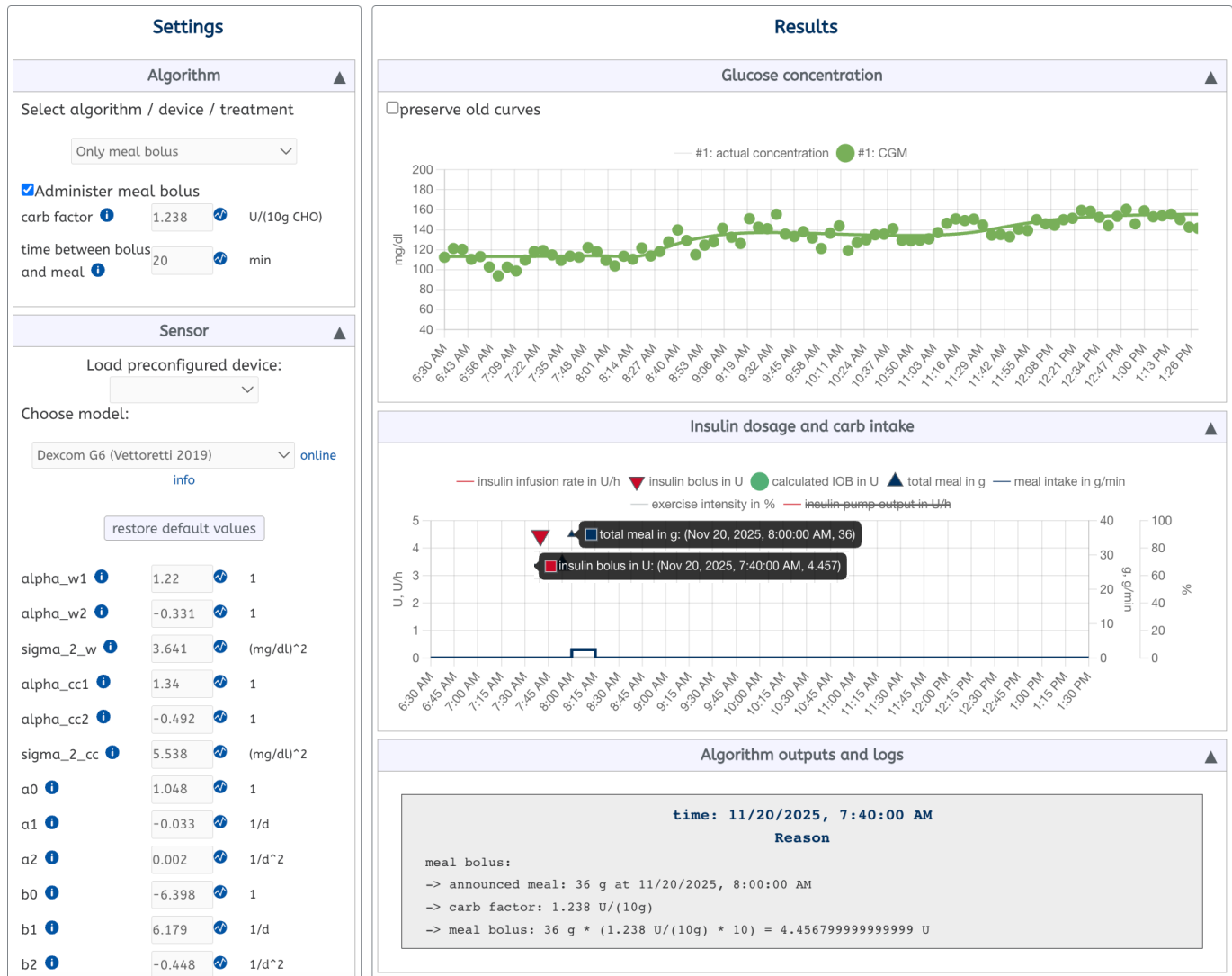


Fig. 3: BGL simulation with LoopInsightT1 simulator for a counterfactual intervention.